問題 318

次の値を求めよ．

解答:

加法定理：$\sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$, $\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta$

(1) $\sin 75^\circ=\sin(45^\circ+30^\circ)=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=\frac{\sqrt{6}+\sqrt{2}}{4}$

(2) $\cos 15^\circ=\cos(45^\circ-30^\circ)=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=\frac{\sqrt{6}+\sqrt{2}}{4}$

(3) $\tan\frac{5\pi}{12}=\tan(75^\circ)=\tan(45^\circ+30^\circ)=\frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}}=\frac{\sqrt{3}+1}{\sqrt{3}-1}=2+\sqrt{3}$

問題 319

α が第 2 象限で $\sin\alpha=\frac{4}{5}$, β が第 1 象限で $\cos\beta=\frac{5}{13}$ のとき，$\sin(\alpha+\beta)$，$\cos(\alpha-\beta)$ を求めよ．

解答:

$\cos\alpha=-\frac{3}{5}$（第 2 象限），$\sin\beta=\frac{12}{13}$（第 1 象限）

$\sin(\alpha+\beta)=\frac{4}{5}\cdot\frac{5}{13}+\left(-\frac{3}{5}\right)\cdot\frac{12}{13}=\frac{20}{65}-\frac{36}{65}=-\frac{16}{65}$

$\cos(\alpha-\beta)=\left(-\frac{3}{5}\right)\cdot\frac{5}{13}+\frac{4}{5}\cdot\frac{12}{13}=-\frac{15}{65}+\frac{48}{65}=\frac{33}{65}$

問題 320

$\sin\alpha=\frac{1}{\sqrt{5}}$, $\sin\beta=\frac{1}{\sqrt{10}}$ (α,β は鋭角) のとき，$\alpha+\beta$ を求めよ．

解答:

$\cos\alpha=\frac{2}{\sqrt{5}}$，$\cos\beta=\frac{3}{\sqrt{10}}$

$\sin(\alpha+\beta)=\frac{1}{\sqrt{5}}\cdot\frac{3}{\sqrt{10}}+\frac{2}{\sqrt{5}}\cdot\frac{1}{\sqrt{10}}=\frac{3}{\sqrt{50}}+\frac{2}{\sqrt{50}}=\frac{5}{5\sqrt{2}}=\frac{1}{\sqrt{2}}$

$\alpha+\beta$ は鋭角の和なので $0<\alpha+\beta<\pi$. $\sin(\alpha+\beta)=\frac{\sqrt{2}}{2}$ より

$\alpha+\beta=\frac{\pi}{4}$ または $\frac{3\pi}{4}$

$\cos(\alpha+\beta)=\frac{2}{\sqrt{5}}\cdot\frac{3}{\sqrt{10}}-\frac{1}{\sqrt{5}}\cdot\frac{1}{\sqrt{10}}=\frac{5}{\sqrt{50}}=\frac{1}{\sqrt{2}}>0$ より

$\alpha+\beta=\frac{\pi}{4}$

問題 321

2 倍角の公式を用いて，次の値を求めよ．

解答:

$\sin 2\alpha=2\sin\alpha\cos\alpha$，$\cos 2\alpha=1-2\sin^2\alpha=2\cos^2\alpha-1$

(1) $\sin\frac{\pi}{8}$

$\cos\frac{\pi}{4}=1-2\sin^2\frac{\pi}{8}$，$\frac{\sqrt{2}}{2}=1-2\sin^2\frac{\pi}{8}$

$\sin^2\frac{\pi}{8}=\frac{2-\sqrt{2}}{4}$，$\sin\frac{\pi}{8}=\frac{\sqrt{2-\sqrt{2}}}{2}$

(2) $\cos\frac{\pi}{8}$

$\cos\frac{\pi}{4}=2\cos^2\frac{\pi}{8}-1$，$\cos^2\frac{\pi}{8}=\frac{2+\sqrt{2}}{4}$

$\cos\frac{\pi}{8}=\frac{\sqrt{2+\sqrt{2}}}{2}$

問題 322

θ が第 3 象限で $\cos\theta=-\frac{3}{4}$ のとき，$\sin 2\theta$，$\cos 2\theta$，$\tan 2\theta$ を求めよ．

解答:

$\sin^2\theta=1-\frac{9}{16}=\frac{7}{16}$. 第 3 象限で $\sin<0$：$\sin\theta=-\frac{\sqrt{7}}{4}$

$\sin 2\theta=2\cdot\left(-\frac{\sqrt{7}}{4}\right)\left(-\frac{3}{4}\right)=\frac{3\sqrt{7}}{8}$

$\cos 2\theta=2\cos^2\theta-1=2\cdot\frac{9}{16}-1=\frac{1}{8}$

$\tan 2\theta=\frac{\sin 2\theta}{\cos 2\theta}=\frac{3\sqrt{7}/8}{1/8}=3\sqrt{7}$

問題 323

三角関数の合成を行え．

解答:

$a\sin\theta+b\cos\theta=\sqrt{a^2+b^2}\sin(\theta+\varphi)$（$\tan\varphi=\frac{b}{a}$）

(1) $\sin\theta+\cos\theta=\sqrt{2}\sin\left(\theta+\frac{\pi}{4}\right)$

(2) $\sin\theta-\cos\theta=\sqrt{2}\sin\left(\theta-\frac{\pi}{4}\right)$

(3) $\sqrt{3}\sin\theta+\cos\theta=2\sin\left(\theta+\frac{\pi}{6}\right)$

(4) $\sin\theta-\sqrt{3}\cos\theta=2\sin\left(\theta-\frac{\pi}{3}\right)$

問題 324

$0\leqq\theta<2\pi$ のとき，$y=\sin\theta+\cos\theta$ の最大値と最小値を求めよ．

解答:

$y=\sqrt{2}\sin\left(\theta+\frac{\pi}{4}\right)$

$-1 \leqq \sin\left(\theta + \dfrac{\pi}{4}\right) \leqq 1$ より

最大値 $\sqrt{2}$（$\theta = \dfrac{\pi}{4}$），最小値 $-\sqrt{2}$（$\theta = \dfrac{5\pi}{4}$）

問題 325

$0 \leqq \theta < 2\pi$ のとき，$\sqrt{3}\sin\theta - \cos\theta = 1$ を解け．

解答:

$\sqrt{3}\sin\theta - \cos\theta = 2\sin\left(\theta - \dfrac{\pi}{6}\right) = 1$

$\sin\left(\theta - \dfrac{\pi}{6}\right) = \dfrac{1}{2}$

$\theta - \dfrac{\pi}{6} = \dfrac{\pi}{6}$，$\dfrac{5\pi}{6}$

$\theta = \dfrac{\pi}{3}$，π

問題 326

次の値を求めよ．

解答:

(1) $\cos\dfrac{5\pi}{12} = \cos(75°) = \cos(45° + 30°) = \dfrac{\sqrt{6} - \sqrt{2}}{4}$

(2) $\sin\dfrac{\pi}{12} = \sin(15°) = \sin(45° - 30°) = \dfrac{\sqrt{6} - \sqrt{2}}{4}$

(3) $\tan\dfrac{7\pi}{12} = \tan(105°) = \tan(60°+45°) = \dfrac{\sqrt{3} + 1}{1 - \sqrt{3}} = -(2 + \sqrt{3})$

問題 327

$\sin\alpha = \dfrac{3}{5}$（$\alpha$ 鋭角），$\cos\beta = -\dfrac{12}{13}$（$\beta$ 鈍角）のとき，$\cos(\alpha + \beta)$ を求めよ．

解答:

$\cos\alpha = \dfrac{4}{5}$，$\sin\beta = \dfrac{5}{13}$

$\cos(\alpha+\beta) = \dfrac{4}{5}\cdot\left(-\dfrac{12}{13}\right) - \dfrac{3}{5}\cdot\dfrac{5}{13} = -\dfrac{48}{65} - \dfrac{15}{65} = -\dfrac{63}{65}$

問題 328

θ が第 1 象限で $\sin\theta = \dfrac{5}{13}$ のとき，$\sin 2\theta$，$\cos 2\theta$ を求めよ．

解答:

$\cos\theta = \dfrac{12}{13}$

$\sin 2\theta = 2\cdot\dfrac{5}{13}\cdot\dfrac{12}{13} = \dfrac{120}{169}$

$\cos 2\theta = 1 - 2\cdot\dfrac{25}{169} = \dfrac{119}{169}$